

AVINDA SHAMAL

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Beliatta, Sri Lanka, 82400

EDUCATION

University of Moratuwa, Sri Lanka

B.Sc. Honours in Electronic and Telecommunications Engineering

May 2022 - Present

CGPA: 3.73/4.00

Expected Graduation: 06/2026

Deans List: 1,4,6,7 Semesters

Selected Coursework: Pattern Recognition, Software Design and Competition, Time Series and Stochastic Processes, Image Processing and Machine Vision, Deep Learning for Vision, Data Structures and Algorithms, Cyber Security in Physical Systems, Introduction to Engineering Optimization, Neural Networks and Fuzzy Logic, Applied Statistics, Linear Algebra, Calculus, Numerical Methods

Ruhunu Vijayaba National College, Beliatta

GCE Advanced Level (Mathematics, Physics, Chemistry)

2011 - 2020
3 As | z score - 2.6852 | Island rank: 242

Certificates

- Deep Learning Specialization - DeepLearning.ai
- Machine Learning - Stanford University
- Train and Manage a ML model with Azure Machine Learning - Microsoft Learn
- AI Agents Fundamentals - Hugging Face
- LSEG CPP Course - LSEG

EXPERIENCE

Fcode Labs [\[Globe\]](#)

Intern AI Engineer

Dec 2024 - May 2025

Nugegoda, Sri Lanka

- Developed and deployed end-to-end AI applications with a focus on secure, reliable, and scalable pipelines, using PyTorch/TensorFlow and GitHub Actions.
- Built an OCR pipeline using CRAFT (text detection) and CRNN (text recognition) models
- Project 1: A client project related to a coffee company
- Project 2: An internal project to generalize image similarity features
- Implemented research paper techniques into production code and optimized existing codebases

SELECTED PROJECTS

Internship Projects

- **AI-Powered Commercial Image Search** | crew.ai, AI Agents [Feb 2025 - Apr 2025]
Contributed to a real-world image search solution for the coffee industry by building AI agent codebases using the CrewAI framework. Managed dataset creation and performed rigorous model evaluations to optimize search accuracy and agent reliability.
- **Generalized Photo Similarity Engine** | JINA, CLIP, Embeddings [Apr 2024 - May 2025]
Conducted internal R&D to generalize AI-based photo similarity features using JINA and local CLIP models. Focused on generating high-dimensional embeddings and benchmarking similarity metrics across diverse datasets to ensure scalability. [\[github\]](#)

Academic Projects

- **Intelligent Knowledge Management Platform(FYP** | RAG, LLMs [Jul 2025] - [Present]
Designed a secure LangChain-based platform featuring a "Dual-RAG" architecture to unify structured and unstructured enterprise data. Implemented dynamic Role-Based Access Control (RBAC), sentiment analysis for user tone detection, and an automated engine to convert natural language into SQL queries and visual analytics. [\[github\]](#)

- **Automated Thermal Inspection System** | *Java Spring, React, AI* [Aug 2025 - Nov 2025]
Developed a full-stack web solution to automate thermal inspection workflows for distribution transformers using Java Spring and React. Integrated an AI-based engine for automatic temperature anomaly detection and interactive tools for generating digital maintenance records. [\[github\]](#)
- **Benchmark for Semantic Bridge Damage Segmentation** | *OpenCV, CNNs* [Sep 2024]
Conducted an image processing and machine vision project focused on segmenting bridge damage for structural analysis. Utilized advanced algorithms to identify and classify damage, contributing to improved infrastructure maintenance and safety. [\[github\]](#)
- **UniVAD Reproduction & Optimization** | *VFM, VLMs* [Aug 2025 - Nov 2025]
Replicated the SOTA UniVAD framework for anomaly detection using Vision Foundation Models and benchmarked performance on the MVTEC-AD dataset. Proposed “Light-UniVAD”, a theoretical optimization framework to compress the heavy architecture into a lightweight U-Net for real-time industrial applications. [\[github\]](#)
- **Jute Pest Classification using Deep Learning** | *DL, CNNs* Mar 2024 - May 2024
Developed a CNN-based solution for agricultural pest identification. Implemented both custom CNN architecture and transfer learning with ResNet50, significantly improving classification performance for jute crop protection applications. [\[github\]](#)
- **IoT Health Tracking System** | *ESP32, PCB, MQTT* [Jan 2024 - Apr 2024]
Developed an IoT-based health monitoring system using ESP32 with MAX30102/MAX30205 sensors to track SpO2, heart rate, and body temperature. Implemented real-time data transmission via MQTT protocol and built a dashboard for remote patient monitoring with alerts. [\[github\]](#)

Personal Projects

- **AI-Powered Document Intelligence System** | *Django, NLP, Celery, Redis* [Dec 2025]
Designed a production-grade Django REST backend for AI-driven document processing, integrating OCR and NLP pipelines. Implemented asynchronous ML workflows using Celery and Redis, built secure semantic search APIs using FAISS and JWT authentication, and dockerized. [\[github\]](#)

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C / C++, MATLAB
- **ML Frameworks:** PyTorch, TensorFlow, OpenCV, Azure ML, LangGraph, LlamaIndex
- **Libraries:** NumPy, Pandas, Matplotlib, Scipy, Sklearn, transformers, datasets, EasyOCR
- **Software and Tools:** Git/GitHub, Docker, VS Code, HuggingFace hub, Linux, PostgreSQL

POSITIONS OF RESPONSIBILITY

Head of Marketing (*Aug 2024 - present*)
SPARK Branch of E-Club, Department of ENTC
University of Moratuwa

Committee Member (*Jan 2024*) - present
Logistic Committee
Mora Maths Society

Assistant Head of Events (*July 2023*)
Industrial Electronic Society of IEEE Student
Branch, University of Moratuwa

Committee Member (*Feb 2023 - present*)
Financial Committee
EXMO 23'

REFERENCES

Dr. Subodha Charles

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Ph.D. (University of Florida, USA), SMIEEE
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Buddhi Vikasitha

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